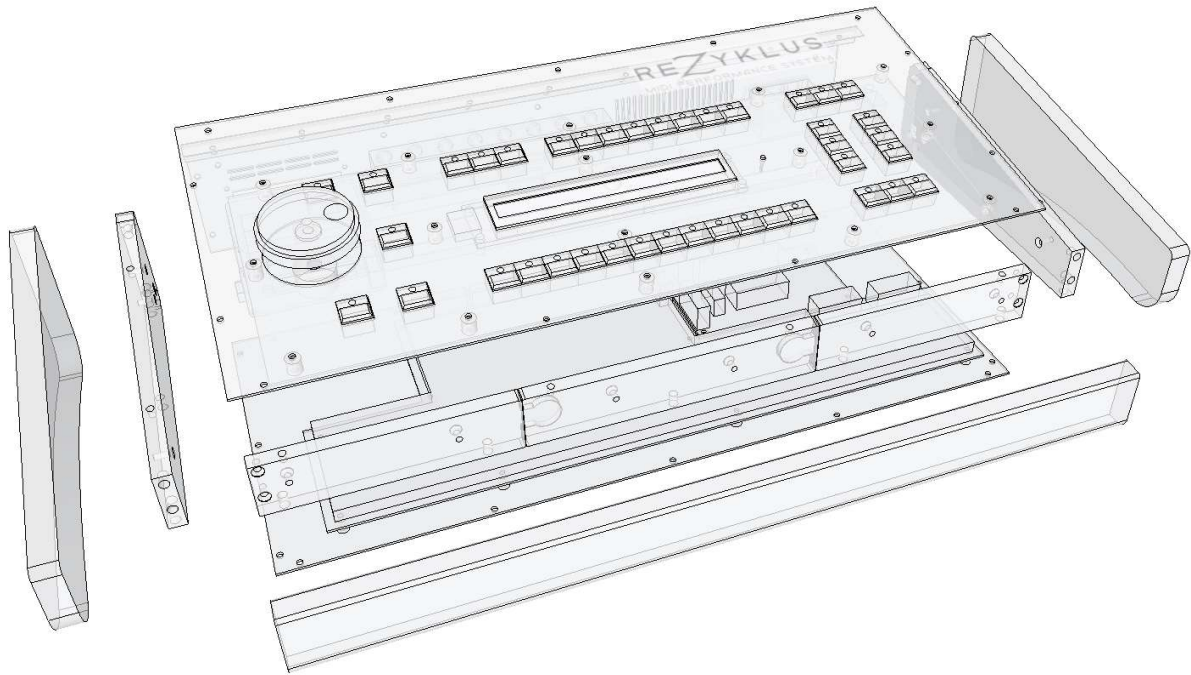
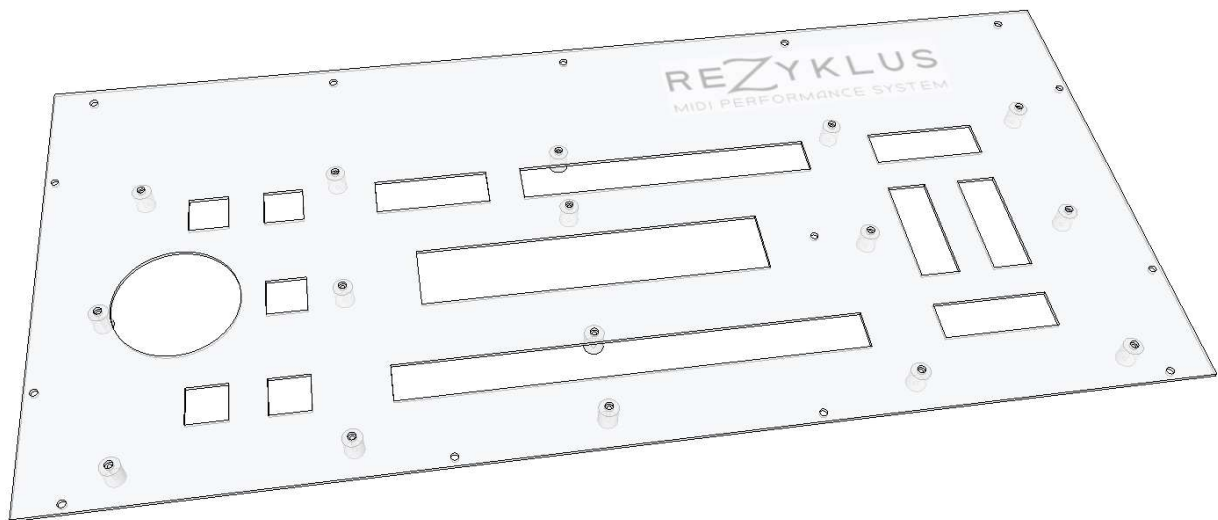




The DIY case for the ReZyklus project is a hybrid of milled & printed aluminium panels (manufactured by JLCPCB), 3D Printed panels (for the sides, front and interconnecting supports) and optionally wooden side and front panels to dress over the 3D printed panels and sharp case edges.

Main Top Panel



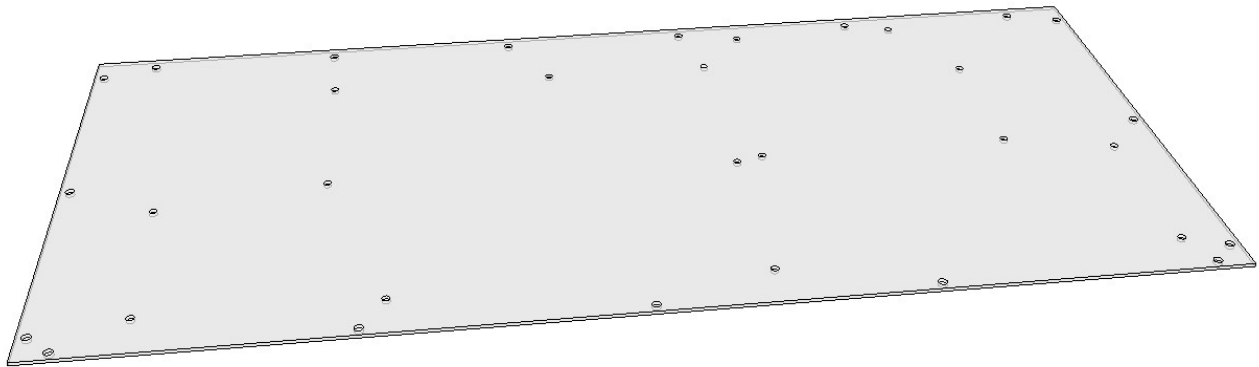
The Panelboard PCB mounts to this using:

- 16x M3 10mm Hexagonal threaded spacers (this size leaves just enough clearance for the LCD)
- 16x M3 6mm button-head Hex bolts (recommended as these are visible on the top panel and smooth to the touch)
- 16x M3 5mm general purpose screws (to secure the PCB to the spacers)

Building the ReZyklus DIY Case

Before fitting the PCB in place, the encoder and Jog Wheel have to be mounted so as to just clear the top of the PCB. Test fit the PCB in place with a couple of screws to check the alignment and if necessary, reposition the encoder pot (may need to file out the mounting hole in the PCB slightly) to get the Jog Wheel central in the panel cut-out.

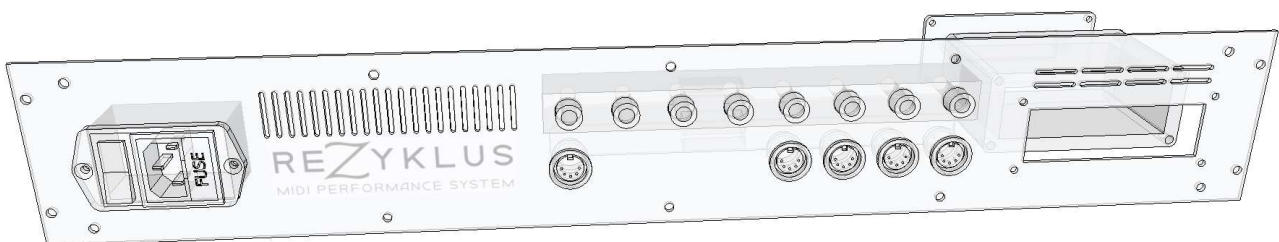
Base Panel



The Mainboard PCB and PSU PCB mounts to this using:

- 11x M3 8mm Hexagonal threaded spacers for the main board (this size leaves just enough clearance for the underside of the Encoder when the case is closed)
- 5x M3 10mm Hexagonal threaded spacers for the PSU board (this allows it to clear the rear mounting strip on the case)
- 16x M3 6mm button-head Hex bolts (recommended as these are smooth to the touch and low-profile on the underside of the case)
- 16x M3 5mm general purpose screws (to secure the PCBs to the spacers)

Rear Panel

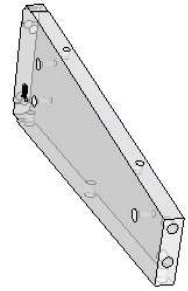


The MIDI Jacks, Input Jacks, IEC Inlet and Cartridge Port attach to this panel using:

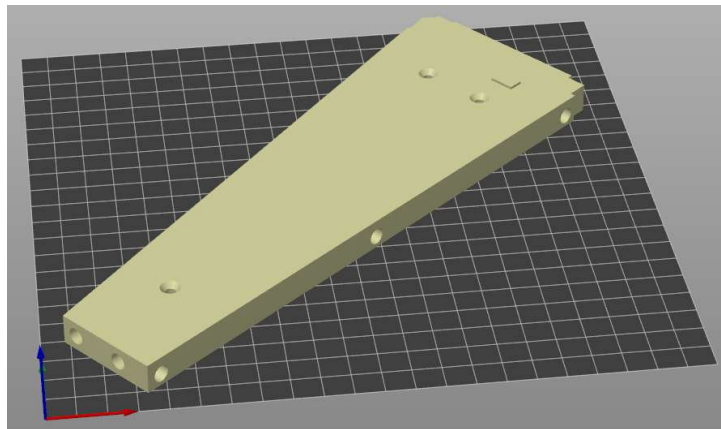
- 2x M3 15mm Countersunk bolts (+ nuts and washers) for mounting the IEC Inlet
- MIDI DIN Sockets – these are mounted to the panel first, before wiring and mounting them to the PCB, as they are mounted from the front of the panel
- Jack Sockets – these can be mounted after soldering to the PCB as they are mounted from the back of the panel
- Cartridge Port – this is a 3D printed part mounted to the case by 4x M3 8mm screws

3D Printed Case Parts

Side Panels

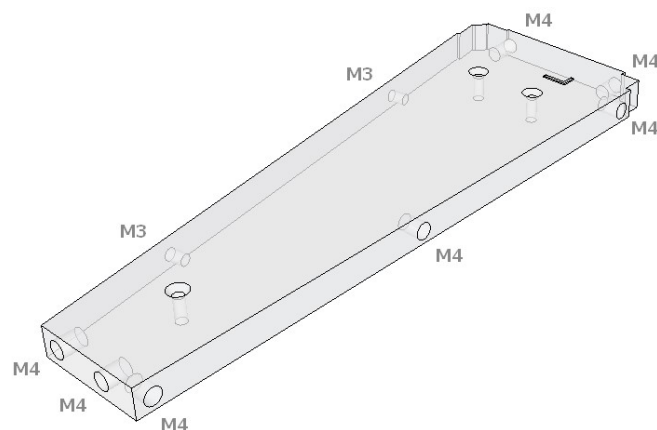


The 3D printed side panels form the main structural parts of the case and have been designed to give the top panel a 10° slope, down towards the front of the case. Each one will fit diagonally on a fairly common print bed size of 230mm x 230mm if printed diagonally:



The orientation should be with the raised letters “L” or “R” on the top surface, and the model needs no support material. Recommended Infill density should be somewhere around 10% Gyroid or 25% Cubic so that plenty of material is printed around the holes where the threaded inserts should be fitted after printing. The panel also includes three countersunk holes for screws to go through for mounting the wooden side panels.

Threaded inserts should be melted into place as per the diagram below (flipped accordingly for the right-hand panel):



Building the ReZyklus DIY Case

A pair of side panels requires a total of:

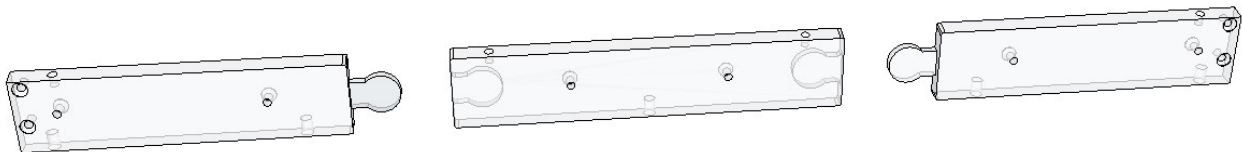
- 4x M3 threaded insert
- 14x M4 threaded insert
- 6x countersunk wood screws (length dependent on the chosen thickness of wood used)

If available, the recommended inserts to use are the longer type with a smooth shaft at one end, as these help keep the insert centrally aligned as they are melted into place:



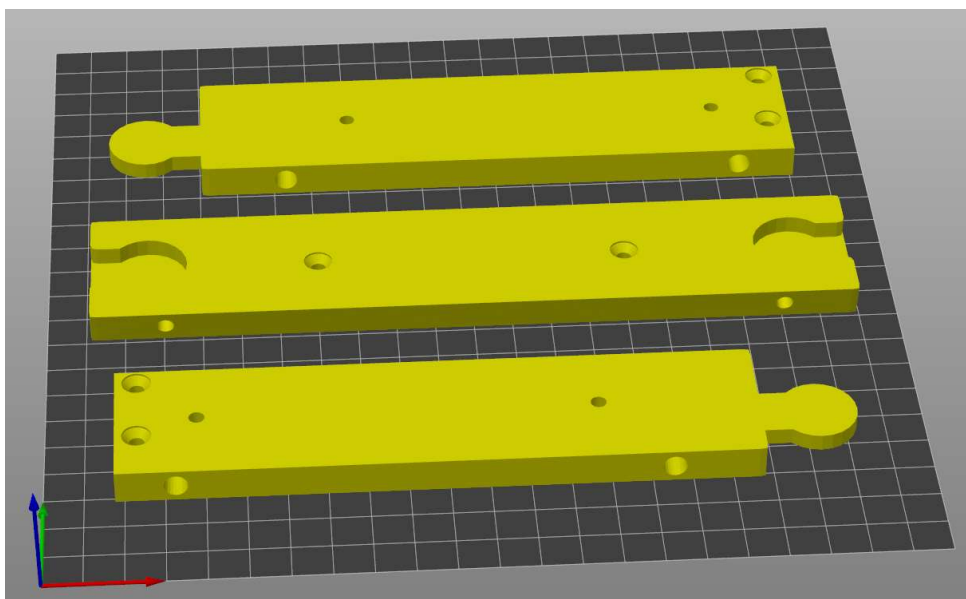
If using a soldering iron to melt the inserts into place, recommend using a conical tip and to set the temperature of the iron to around 220°C. As soon as the insert is in place, thread an M3 bolt into the insert and use it to adjust the inserts alignment and draw the heat out quickly.

Front Panels



The 3D printed front panels form part of the structural parts of the case and have been designed to match the top panel 10° slope. The panel is split into three pieces in order to fit on a standard print bed, however all three can be printed at once.

Note that the Left and Right sections should be printed with the tabs touching the print bed, but the center section oriented with the tab cut-outs on the top surface:



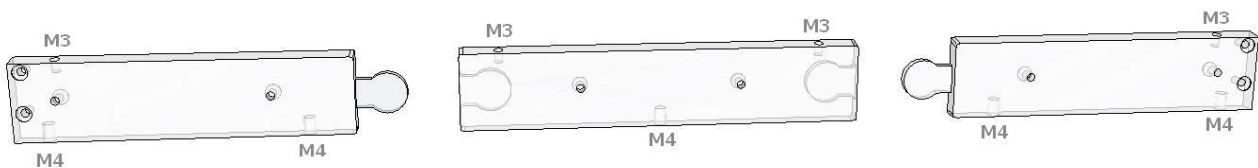
Building the ReZyklus DIY Case

The models need no support material and, as with the other parts, the recommended Infill density should be somewhere around 10% Gyroid or 25% Cubic so that plenty of material is printed around the holes where the threaded inserts should be fitted after printing.

The panels also include six countersunk holes for screws to go through for mounting the wooden front panel. There are also two countersunk holes at each end for securing the front panels to the side panels via 4x M4 12mm Countersunk bolts.

The tabs on the end sections are designed to be glued into place in the center section (an adhesive compatible with the chosen filament is recommended, but CA glue will usually suffice). When doing this, it is recommended to temporarily mount the panels to the base plate so as to ensure the correct spacing and alignment, then “wick” the CA glue around the tabs.

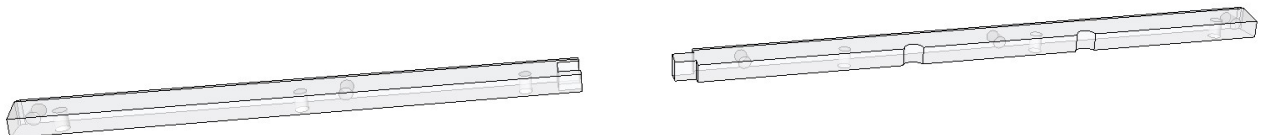
Threaded inserts should be melted into place as per the diagram below:



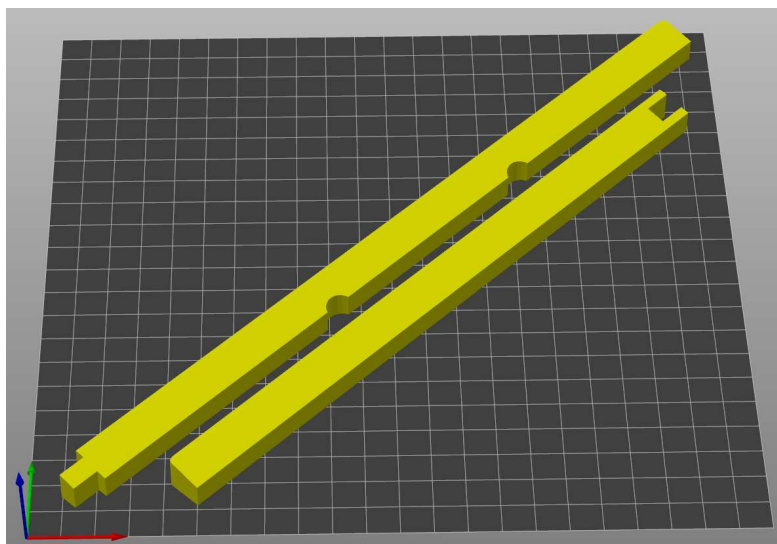
A set of three front panels requires a total of:

- 4x M3 threaded inserts
- 5x M4 threaded insert
- 4x M4 12mm Countersunk bolts
- 6x countersunk wood screws (length dependent on the chosen thickness of wood used)

Rear to Base Panel Mounting Strip



This part ensures the rear panel and base plate are mounted rigidly to each other. It is split into two sections that can be printed diagonally on a standard print bed:



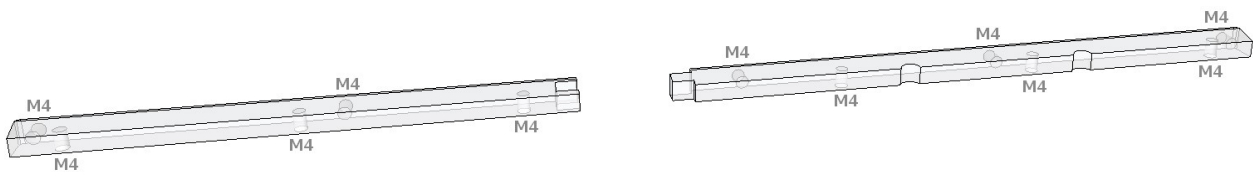
Building the ReZyklus DIY Case

The models should be printed with the insert holes oriented face down onto the build plate. The models need no support material, and the recommended Infill density should be somewhere around 10% Gyroid or 25% Cubic so that plenty of material is printed around the holes where the threaded inserts should be fitted after printing.

The set of two strips require a total of:

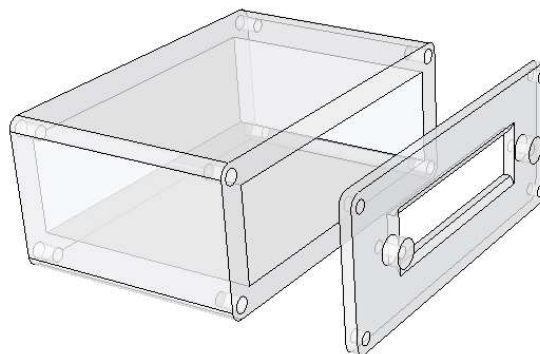
- 11x M4 threaded inserts
- 11x M4 6mm button-head Hex bolts (recommended as these are smooth to the touch and low-profile on the case panels)

Threaded inserts should be melted into place as per the diagram below:



The center parts of the models are designed to fit into each other and can also be glued together if desired, although it is not necessary.

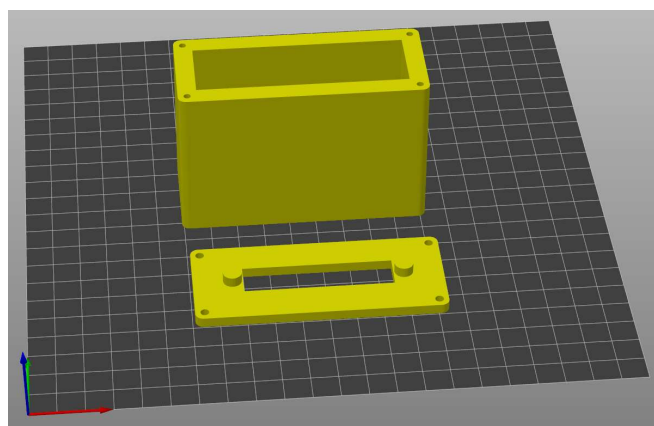
Cartridge Port



This consists of two part that screw together via M3 bolts and M3 threaded inserts:

- The sleeve to guide the cartridge into place
- The connector plate to mount the 37-way Male IDC connector into

Both models fit easily onto a standard build plate. The sleeve should be printed upright, and the connector plate printed flat to the bed:



Building the ReZyklus DIY Case

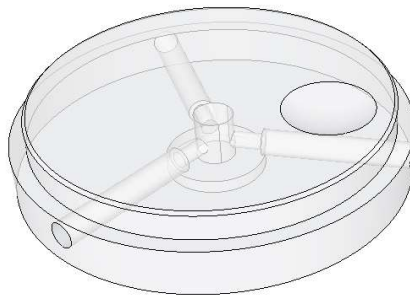
The models need no support material, and the recommended Infill density should be somewhere around 10% Gyroid or 25% Cubic.

The two parts require:

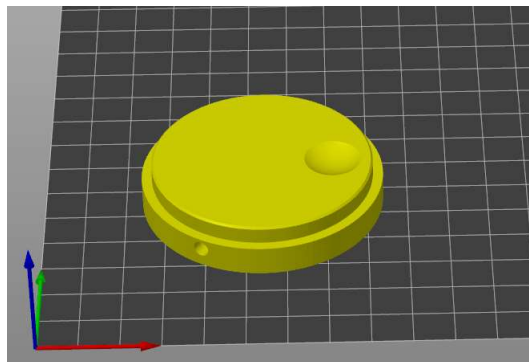
- 8x M3 threaded inserts
- 8x M3 8mm self-tapping screws (4x to attach the connector plate to the sleeve, and 4x to mount the sleeve to the rear panel)
- 2x M3 6-10mm self-tapping screws to mount the connector to the plate; the length of these screws will depend on whether the connector is metal-bodied or plastic-bodied.

The connector should be mounted such that it is on the inside of cartridge port, rather than from the back. (It is easier to first mount the connector to the plate, then mount the whole assembly to the sleeve)

Encoder Jog Wheel



This part is modelled on the original Zyklus MPS aluminium wheel. It needs a little work after printing to drill through and tap the three grub-screw mounts. Print the part oriented as below:



Recommendation is a dense infill (e.g. 20% Gyroid or 50% Cubic), 2 perimeters and at least 4 layers to get a smooth finish on the top. Layer height should be at least 0.20mm, but if the “dimple” in the top is rough, try again with 0.16mm instead. The part will also need support on the base plate for the recessed center section underneath.

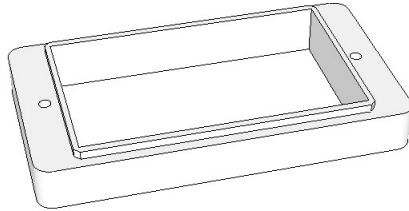
After printing, first remove the support material. Carefully drill the three grub screw channels with a 2mm drill bit, through to the center opening where the shaft goes in. The ends of these channels (nearest the center) will then need threading with an M3 tap, after which 3x M3 grub screws will need inserting; use at least 8mm screws to get good alignment and purchase on the threads.

The wheel should fit tightly on the encoder shaft, the grub screws helping with alignment and eliminating any wobble when turning the wheel. It can take some time to get this right, but it is possible.

Building the ReZyklus DIY Case

An alternative would be to have this part Resin Printed for a better finish, however it may be necessary to alter the STL model to accommodate the different printing process.

IEC Mains Inlet Mount



This part is intended to act as a “backing plate” for the combination IEC Inlet and Power switch, specifically part: **Bulgin BVA01/Z0000/02**

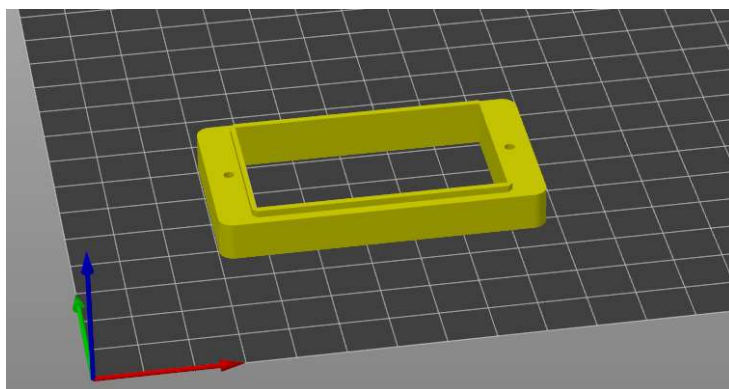
- /02 is the suffix for a red neon-lit switch
- /01 is the suffix for a black non-lit switch



However, this part is not supplied as a moulded one-piece design; instead it consists of a power switch and IEC inlet snapped into a plastic bezel and is not particularly well made (despite the brand and price!). It requires some hot glue or epoxy to secure the parts in the mounting bezel.

The part mounts from the front-side of the panel, and the 3D printed backing plate helps with alignment and also provides holes for 2x M3 8mm Self-tapping screws.

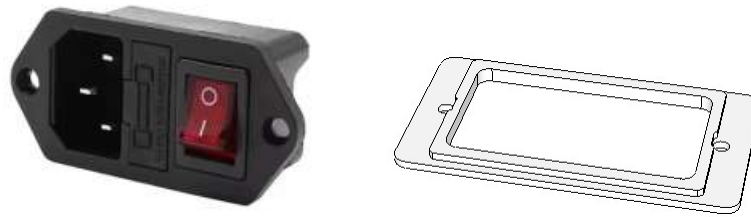
The part should be printed flat onto the print bed:



Recommended infill is 20% Gyroid or Rectilinear, just to provide plenty of material around the mounting holes, 2 perimeters and 3 top & bottom layers (pretty standard settings).

Building the ReZyklus DIY Case

UPDATE: There are better inlets available (AC-01A) which are one-piece designs, however these require a slightly different sized cut-out on the panel. To accommodate these, a 3D printed insert plate has been designed to fill in the gaps around the inlet:

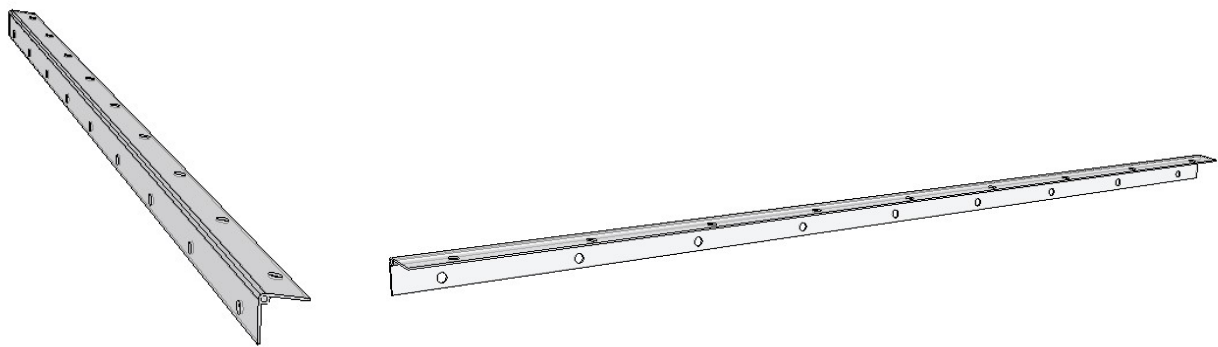


This would be an easy print, flat face down on the print bed with 100% infill.

Note that when fitting this type of inlet, the mounting holes on the case panel would need to be filed into slots to allow for the closer spacing of the mounting holes on this type of inlet.

Hinge Assembly

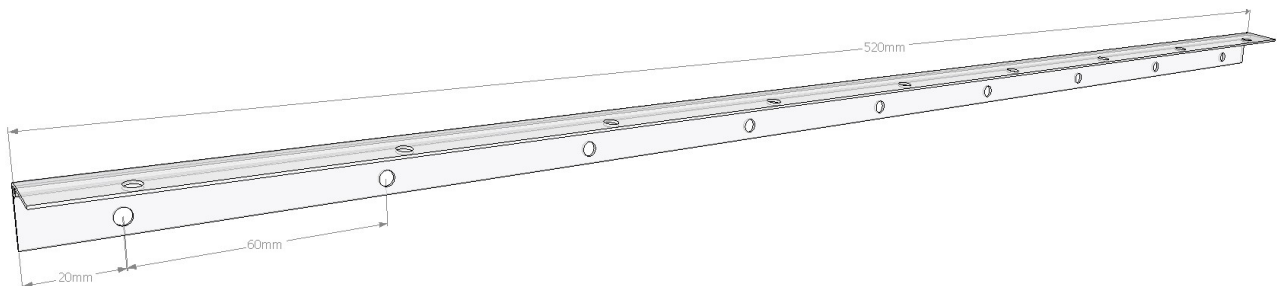
(shown below in two orientations)



The DIY case was designed with a hinge mounting the top and rear panels together; this provides a neat solution to the 80° angle between the two and allows for easy opening to get inside the unit.

The brass hinge used was obtained from a local hardware store (Wickes in the UK) but seems to be pretty standard in dimensions/hole spacing as those found on sites like Amazon.

It will require cutting to length, probably from both ends to ensure the holes line up with the panel. The cuts should be done with either a rotary tool (e.g. Dremel) cutting disc, or a fine bladed hacksaw. The length and mounting hole spacing should be as shown below:



The hinge is attached on the inside of the two panels using M4 8mm Button-head Hex bolts and suitable nuts & washers (suggest Nyloc type nuts); note that not every hole on the hinge is used.

Document Revision History

Date	Version	Notes
13/10/2024	v0.1	Initial Draft
14/10/2024	v0.2	Minor corrections
11/12/2024	v0.3	PSU PCB requires 10mm spacers. Updated some images.